

- 3 Stomata have daily rhythms of opening and closing. Stomatal aperture width and the concentration of potassium ions in guard cells were measured in the leaves of basil plants, *Ocimum basilicum*. The basil plants were grown in conditions of natural light, with optimum temperature and soil moisture.

Figure 3.1 shows the results obtained between the times of 06:00 and 17:00. Error bars have been included.

key

- mean stomatal aperture width
 --■-- concentration of potassium ions in guard cells

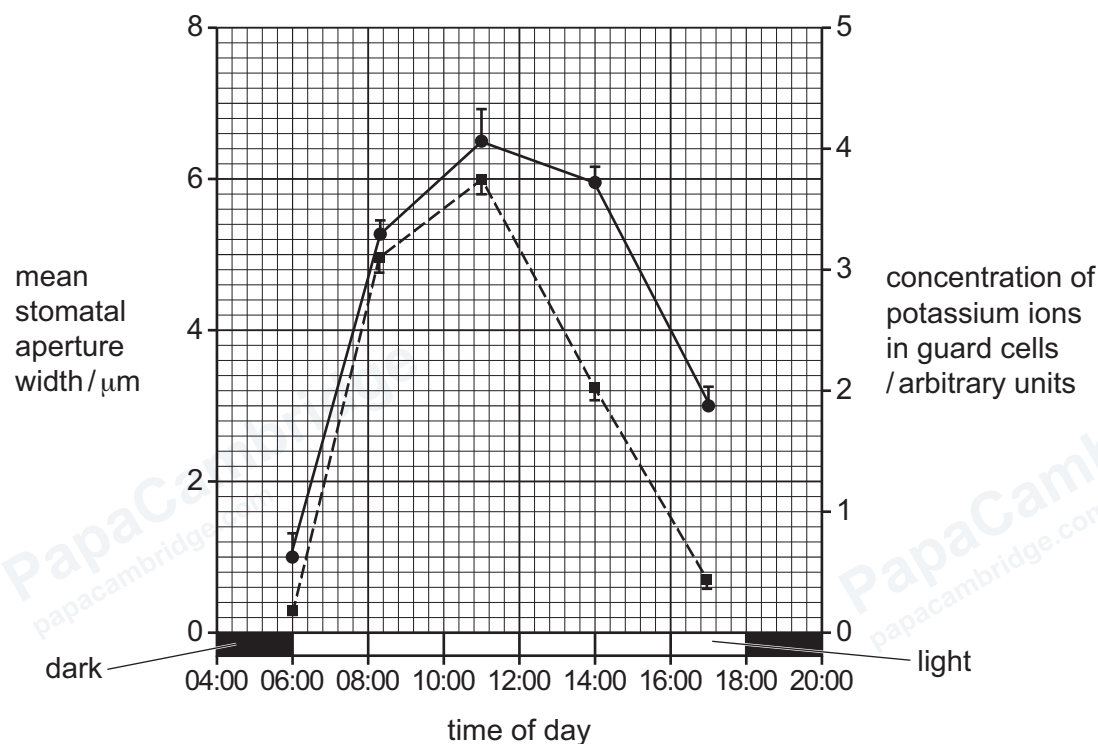


Figure 3.1

- (a) (i) With reference to Figure 3.1, describe the changes that occur in mean stomatal aperture width **and** suggest reasons why these changes benefit basil plants.

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- (ii) Figure 3.1 shows there is a correlation between mean stomatal aperture width and the concentration of potassium ions in guard cells.

Explain how a change in one of these factors may cause a change in the other factor.

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- (b) Researchers observed a daily rhythm in the synthesis of abscisic acid (ABA) in plants.

Suggest whether ABA concentration is more likely to increase when light intensity is increasing **or** is more likely to increase when light intensity is decreasing.

Give reasons for your answer.

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